

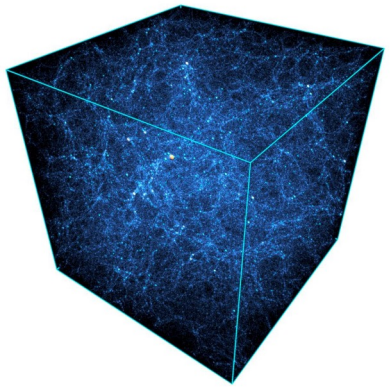
Section 1: Why lossy compression for scientific data and use-cases

Franck Cappello, Argonne National Laboratory and the University of Chicago

Why Lossy Compression

- Motivations,
- General principles
- The role of Shannon Entropy
- Lossy compression principles
- Use-cases

The Flood of Scientific Data!



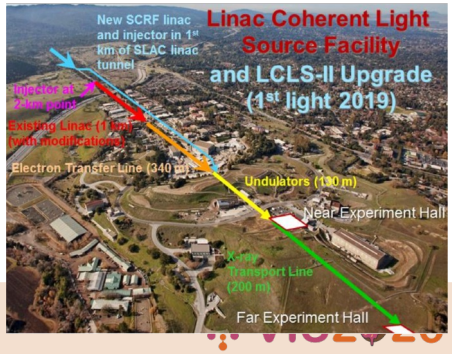
https://www.anl.gov/article/big-bang-the-movie

- **Today's scientific research is using simulation or instrume and produces extremely large of datasets to process/analy**

- Cosmology Simulation:
  - A total of >**20PB** of data when simulating trillion of particles
  - Petascale systems FS ~20PB (you will never have 20PB of scratch for one application)
- Climate Simulation:
  - High-resolution CESM simulation (1/10° ocean-ice, 1/4° atmosphere)
    → 1 TB of data generated per compute day
  - NCAR platform (2013): ~20% of hardware budget
  - NCAR platform (2017): ~50% of hardware budget
- High-energy x-ray beams experiments
  - LCLS (Linac coherent light source) upgrade

  → **1TB/s of streaming intensity**

  → X1000s of disks



T.O. Raubenheimer for the LCLS-II Collaboration, SLAC, Menlo Park, CA 94025, USA
6th International Particle Accelerator Conference, VA, USA, 2015, JACoW Publishing

Data compression

- A form of data reduction
- ~50 years old (LZ77), ~80 years (Shannon's information theory - 1948)
- Systematically (almost) used for files (e.g. GZIP), digital photos (e.g. JPEG), movies (e.g. MP4), music (e.g. MP3)
- Very effective on images, videos.
- A priori, compression algorithms are generic and applicable to many applications
  - In practice they are combined and optimized for specific usages

Compression and information loss

- **Lossless**:
  - The data is compressed and decompressed in such a way that the decompressed data is **identical** to the initial data (there is no alteration, deviation, distortion)
  - Popular example: Gzip, Zstandard

- **Lossy**:
  - The data is altered during compression: some piece of information is removed (lost): **the original data cannot be retrieved**.
  - Noise is added + potential distortions to the original data. Knowing the nature of the noise and distortions is critical for users of the compressed data.
  - If user can control the accuracy of the compressed data, then lossy compression is a trade-off between compression ratio and loss of accuracy
  - Popular example: JPEG, MP3

Role of the Shannon Entropy

- Is a measure of *unpredictability* of *information content*.
- Considers the set of symbols in a string and their probability of occurrence (not only diversity!)
- Computes how many digits (in average) are needed to code each symbol of the string

- Entropy of the string: $$H = -\sum p(x) \log p(x)$$

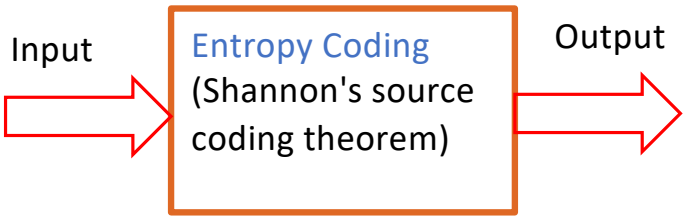
If p(x) is high (common event), log p(x) is less negative, so -log p(x) is small (less surprising)
If p(x) is low (rare event), log p(x) is very negative, so -log p(x) is large (more surprising)

- p(x) being the probability of occurrence of x in the string
- -log p(x): information quantity carried by observing x
- $\log_b$ in this formula. For binary digits, b=2. For decimal digits, b=10

When b=2, H is the number of bits in average to code each symbol of the string

- H high → X distribution is uniform (similar probability of occurrence for each symbol)
- H low → X distribution is varied (some symbols are more frequent than others)

Directly Encoding Scientific Datasets?



Each floating-point number is a symbol

Shannon Entropy of a string of symbols:

$$H = -\sum p(x) \log p(x)$$

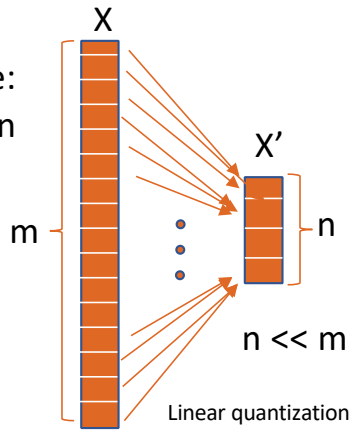
p(x) being the probability of occurrence of x in the string
log p(x): information quantity carried by observing x
(the closer p(x) is to 1, the less information, observing x Once again does not carry new information)

Notes (for the 32 bit case):
-iavuh: If all values were unique in the dataset, h would equal to:
-With entropy at 24-25, max compression will be 32/25= 1.28
-*all fields converted to 32bits for uniformity in the table

Fields		8bit_entropy	32bit_entropy
Hurricane Isabel (Fluid Dynamics)			**iavuh=24.5**
100x500x500 32 bits	CLOUDf48	1.373796	3.012497
	Pf48	7.29214	24.126759
	PRECIPf48	7.469055	24.066132
	QCLOUDf48	0.66233	1.306124
	QGRAUPf48	7.422313	23.383468
	QICEf48	1.140779	2.424523
	QRAINf48	7.408208	23.120446
	QSNOWf48	7.447305	23.64042
	QVAPORf48	7.389244	24.187435
	TCf48	7.197273	23.542626
	Uf48	7.358883	24.179438
	Vf48	7.352912	24.182481
	Wf48	7.40205	24.238407
HACC (Cosmology)			**iavuh=30**
1B particles 32 bits	vx	7.370827	25.947039
	vy	7.366939	25.923527
	vz	7.367481	25.916744
	x	7.120996	24.820431
	y	7.112814	24.77008
	z	7.100023	24.713558
Miranda (Radiation hydrodymanics)			**iavuh=25.16**
256x384x384 32 bits*	density	3.620953	7.224017
	diffusivity	7.722102	25.052816
	pressure	7.723669	25.074806
	velocityx	7.558891	24.97001
	velocityy	7.563981	24.974097
	velocityz	7.583583	24.985736
	viscocity	7.718522	25.049896

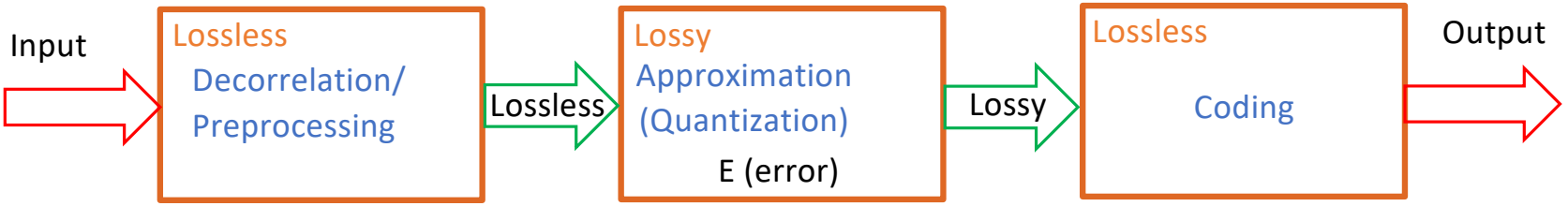
Beyond compression ratio of 2: Lossy Compression

Example:
Linear quantization



Use the minimum number of bits to represent a string of symbols

Exploit Redundancy



Where the compression error is introduced (e.g. removing coefficients of a transformed matrix)

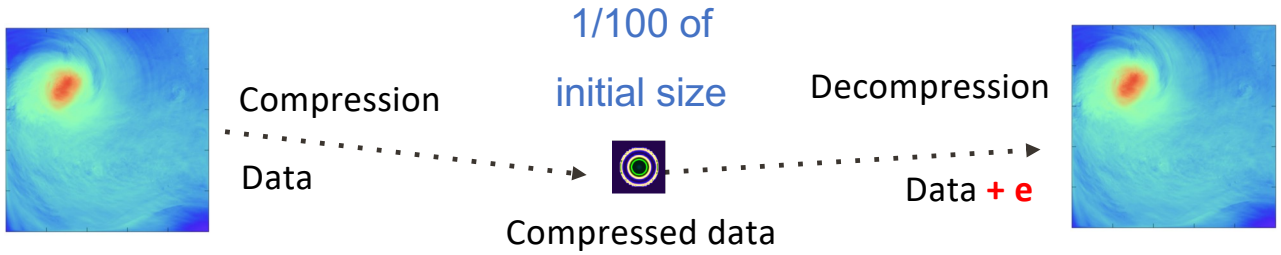
Lossy compression requirements

- Reduces data accuracy to reach higher compression ratios (typically 1 order of magnitude or more) than lossless compression.
  - If lossy compression does not provide higher compression ratios → use lossless.
- Tradeoff between data size and data accuracy

- Specific requirements for usefulness:

We will define accuracy in later in the tutorial

  - Error-bounded compression: **guaranteeing** the ~~accuracy~~ of the decompressed data for users
  - **Fast** compression and decompression (if in-situ, compression time should be much lower than storage time)

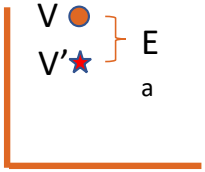


Images from: X. Liang, S. Di, D. Tao, S. Li, S. Li, H. Guo, Z. Chen, F. Cappello, Error-Controlled Lossy Compression Optimized for High Compression Ratios of Scientific Datasets, **IEEE BigData** 2018

Basic compression error bounds

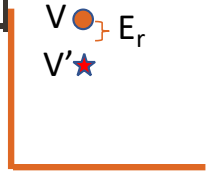
- Advanced lossy compressors (SZ, ZFP) provides **point wise** error controls (bounds)
- Two main types of error bounds (E: error bound, V: initial dataset, V': decompressed dataset):
  - Absolute error bound: (absolute quantities. E.g. position in space),

$$E_a = |V - V'|$$



  - Relative error error bound: (e.g. velocity, temperatures),

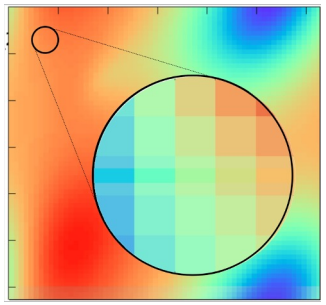
$$E_r = \frac{|V - V'|}{V}$$



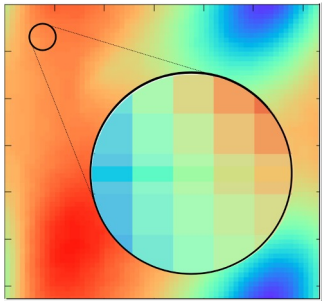
Remark 1: Lossy compression produces high quality data

ECP QMCPack:
Quantum Monte Carlo simulation
3d cubic spline coefficients of Single-Particle Orbitals

Original

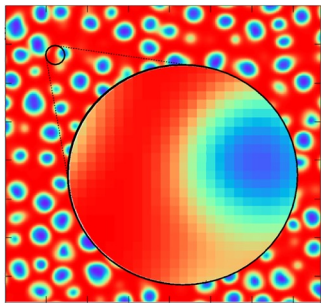


SZ Compression ratio: 8.8
PSNR=120dB
Signal/error ~10^6

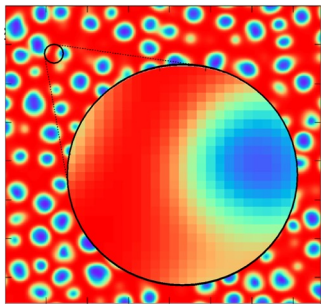


Miranda:
Radiation hydrodynamics simulation
Density field

Original

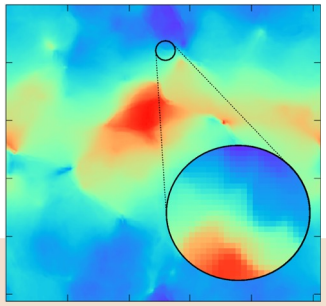


SZ Compression ratio: 30
PSNR=96dB
Signal/error ~10^5

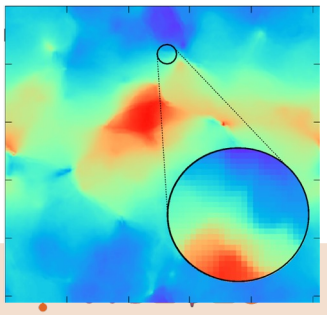


NYX:
Cosmology simulation
X velocity field

Original



SZ Compression ratio: 143
PSNR=64dB
Signal/error ~10^3



Some use-cases of lossy compression

We are seeing an increasing diversity/number of use-cases

"Classic" use-cases:

1) Visualization
2) Reducing storage footprint (offline compression)
3) Reducing I/O time (on-line, in-situ compression)

Recently identified use-cases:

4) Reducing streaming intensity (recent for generic floating-point compressors)
5) Lossy checkpoint/restart from lossy state
   - reduce checkpoints footprint on storage – adjoint, accelerate checkpointing
6) Avoiding re-computation
   - By reducing the memory footprint
7) Running larger simulations
   - By reducing the memory footprint
8) Accelerate computation
   - By reducing the memory-byte/flop ratio (reducing the pressure on memory bandwidth) → see H. Fu, et al., "18.9-Pflops Nonlinear Earthquake Simulation on Sunway TaihuLight: Enabling Depiction of 18-Hz and 8-Meter Scenarios", Proceedings of SC17
9) DNN model/activation map/training set compression
10) Compression of MPI communications

Goals:

- Show the diversity of uses cases through real-life applications
- Present real-life performance gains
- Inspire potential new use-cases

Source:
Most examples from the Exascale Computing Project (ECP).

Most examples use the SZ compressor.

Reducing Storage Footprint and Accelerating I/O 1/2

ECP HACC: N-body problem with domain decomposition, medium/long-range force solver (particle-mesh method), short-range force solver (particle-particle/ particle-mesh algorithm).

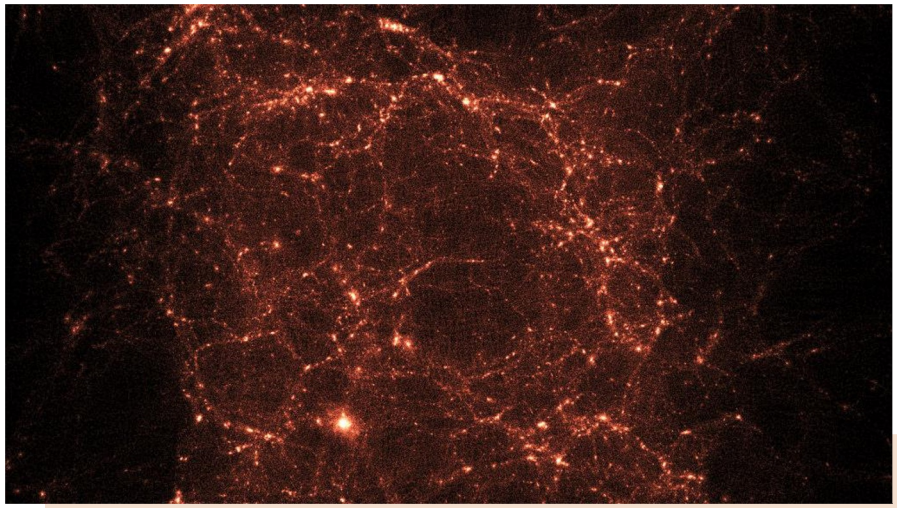
Particle dataset: 6 x 1D array (x, y, z, vx, vy ,vz)

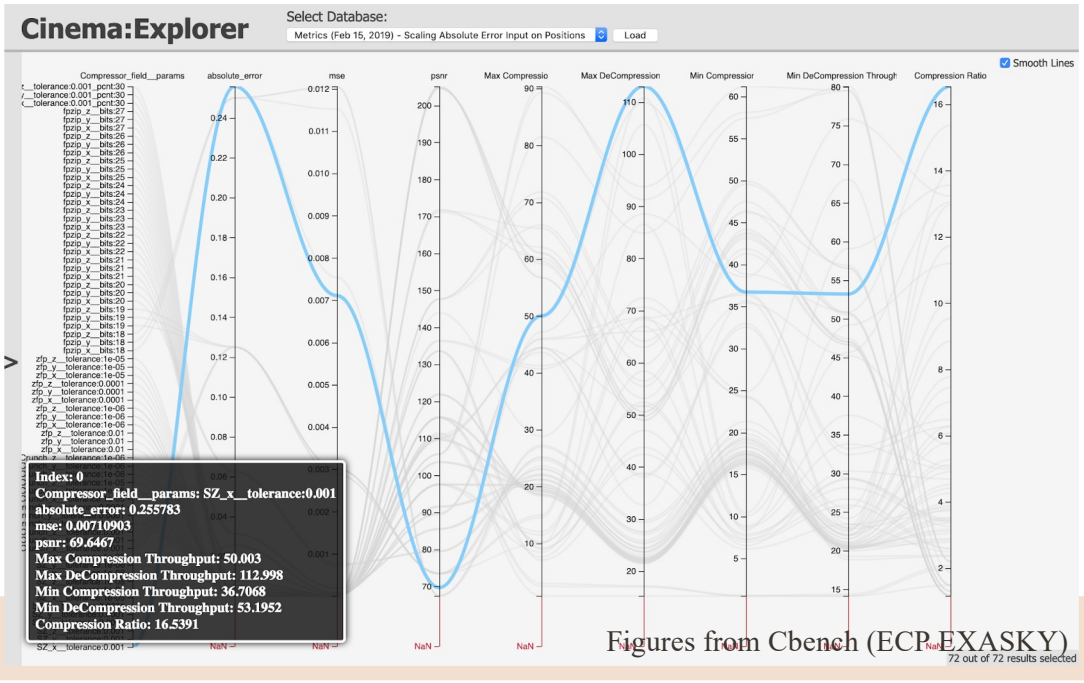
SZ 2.0: **CR ~5** (~6bits/value) at 10^{-3} error bound

Preferred error controls:

- Point wise max error bound
- Absolute (position), Relative (Velocity)

ANL: Cosmological Simulations for Large-Scale Sky Surveys



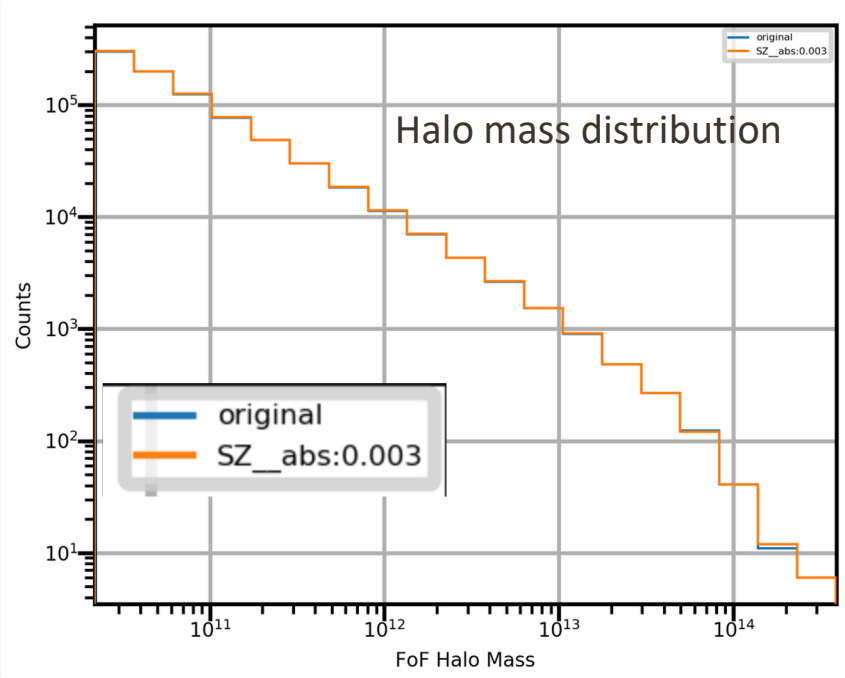


Figures from Cbench (ECP EXASKY)

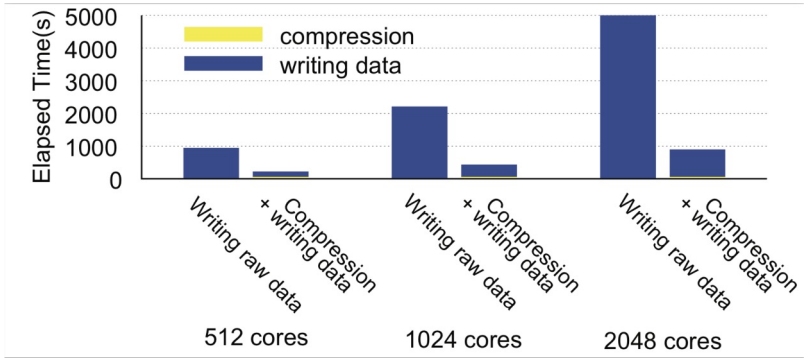
Reducing Storage Footprint and Accelerating I/O 1/2

ECP HACC Results validation

3 kpc absolute error

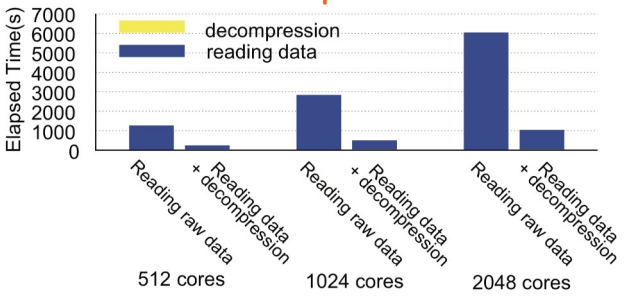


Compression



- Bebop cluster at Argonne GPFS file system
- File-per-process mode with POSIX I/O for reading/writing data in //
- Each core has 3.14GB to write

Decompression



I/O time reduced by ~5x-~6x

The writing and reading times without compression are estimations (too long for experiment)

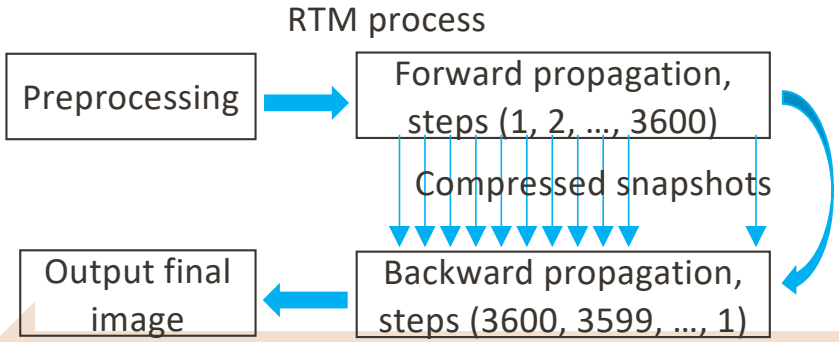
S. Li, S. Di, X. Liang, Z. Chen, F. Cappello, Optimizing Lossy Compression with Adjacent Snapshots for N-body Simulation Data, **IEEE BigData 2018**

Reduction of memory footprint: Seismic Imaging 1/2

Reverse Time Migration (RTM) is a commonly used method by oil companies for seismic imaging in areas with complex geological structures.

The RTM process involves forward propagation (wave simulation) storing snapshots (3600 in the presented example) and a backward propagation using these snapshots to compute the final image.

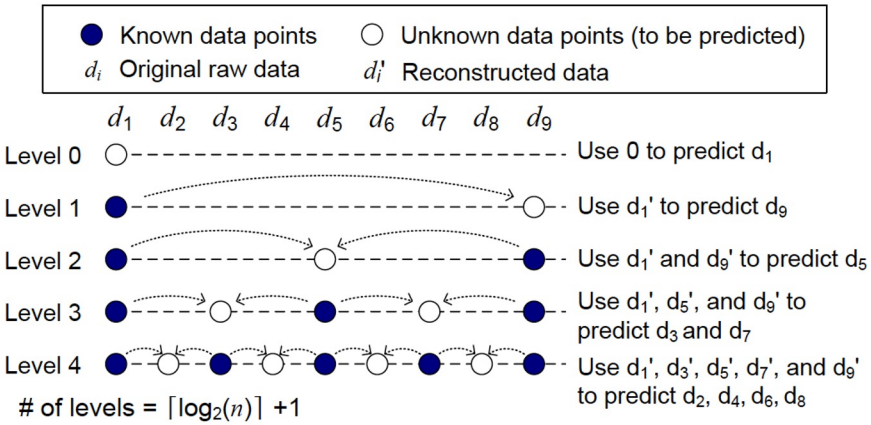
Compression use value-range relative error bound of 10^{-3} because the value ranges of the snapshots are different.



SZ Interp: Predictor based on multilevel, multidimensional tri-cubic spline interpolation

Spline method	*Prediction Value* p_i
Linear spline	$p_i = \frac{1}{2}d_{i-1} + \frac{1}{2}d_{i+1}$
Cubic spline	$p_i = -\frac{1}{16}d_{i-3} + \frac{9}{16}d_{i-1} + \frac{9}{16}d_{i+1} - \frac{1}{16}d_{i+3}$

1D case (linear spline):



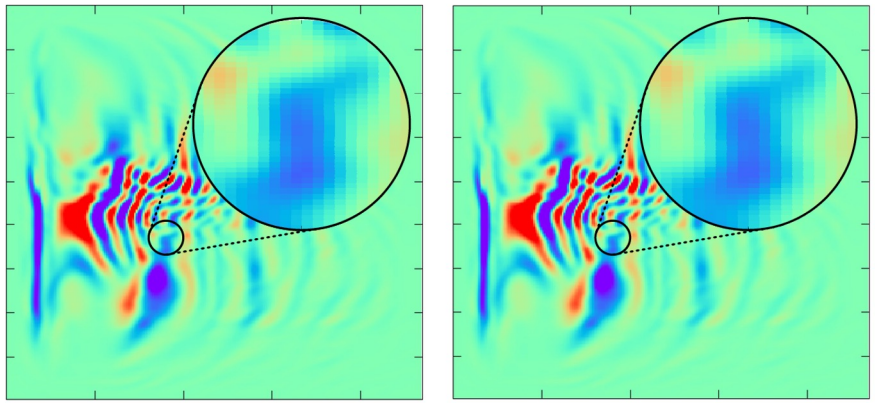
At level 0, 0 to predict $d1$, □ Store quantized error (e0)
At level 1, $d1$+e0 to predict $d9$, □ Store quantized error (e9)
At level 2, $d1$+e0 and $d9$+e9 to predict $d5$, □ Store error (e5)
…

Reduction of memory footprint: Seismic Imaging 2/2

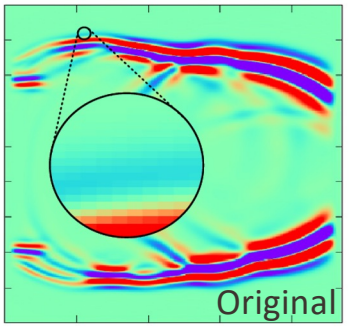
Whole simulation final image
after 3600 time steps

CR=274

Final image from Reverse Time Migration

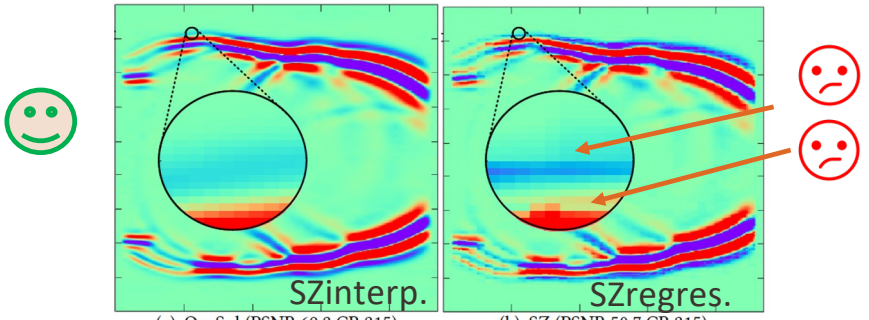


(a) Original Final Result (b) Compression-based Final Result

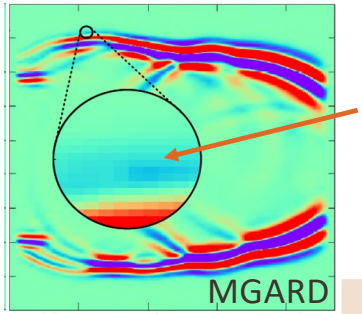


CR:~315

Figure 25: Visualization of RTM, original data



(a) OurSol (PSNR:69.3,CR:315) (b) SZ (PSNR:50.7,CR:315)



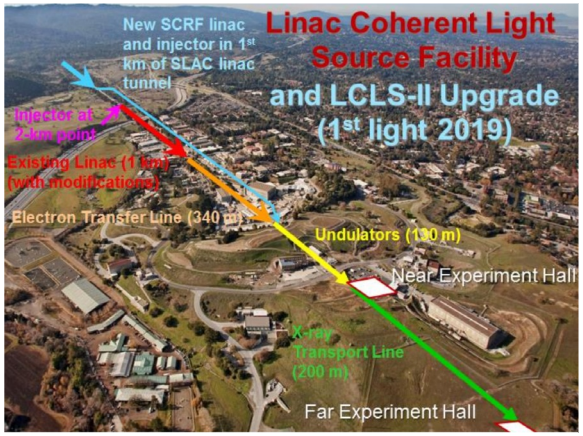
(d) MGARDx (PSNR:62.5,CR:310)

With this level of compression, the RTM process can be executed without I/O

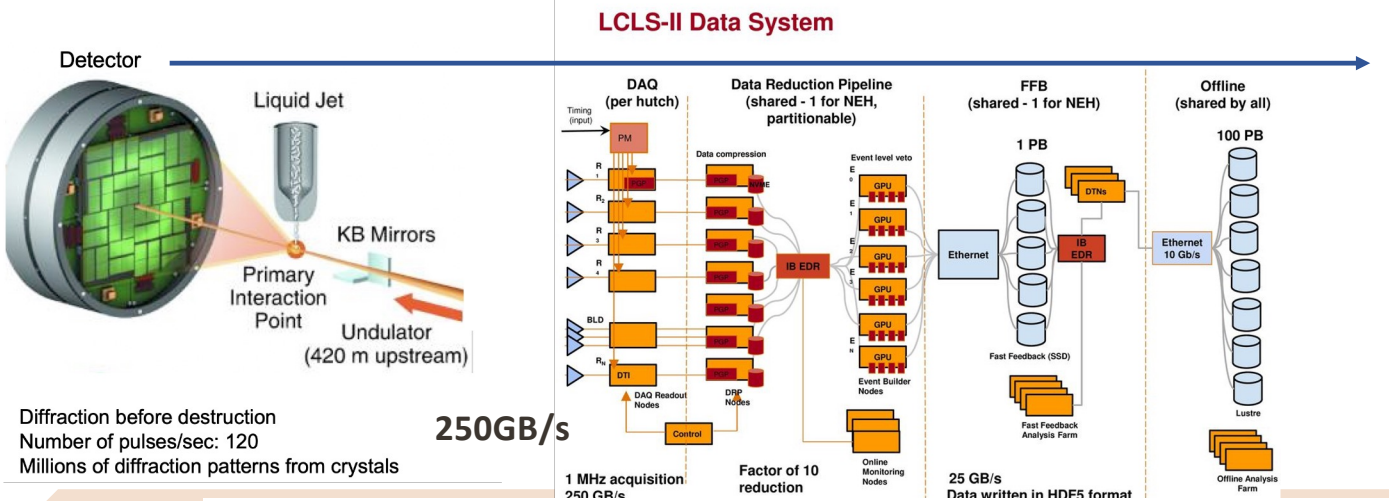
Reducing Streaming Intensity 1/3

ECP EXAFEL: Context of LCLS-II

- X-ray crystallography to determine the atomic and molecular structure of a crystal
- Crucial for medical applications (e.g. understand membrane protein structure)
- A liquid jet injects the crystal and an Xray pulse strikes the molecules.
- This produces diffraction patterns on the detector
- Patterns are analyzed to discover the structure of the crystal/molecule



T.O. Raubenheimer for the LCLS-II Collaboration, SLAC, Menlo Park, CA 94025, USA
6th International Particle Accelerator Conference, VA, USA, 2015, JACoW Publishing



Images from: LCLS 2 Data Reduction Pipeline Preparations for Serial Femtosecond Crystallography Chuck Yoon HDRMX, Mar 16, 2017

Reducing Streaming Intensity 2/3

-Detector produces 2D images:

-4M pixel/event

-All LCLS-II area detectors per experiment: 250GB/s

-Today technology: ~x1000s disks to sustain the bandwidth

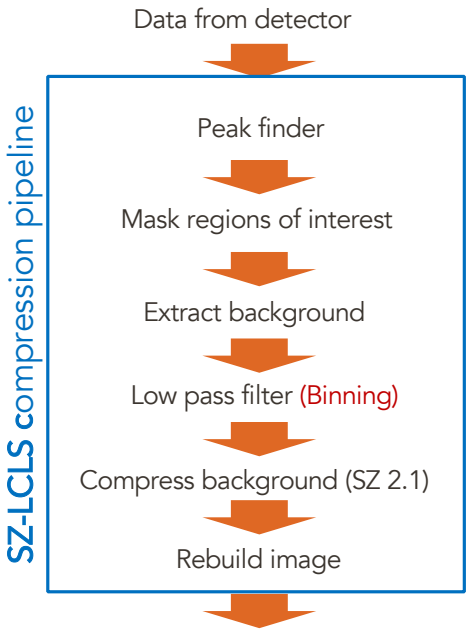
-Data is unsigned integers (RAW, Calibrated), in binary XTC2 format

Compression goals:

-**Goal: CR of 10** with error bound

-Speed 500 MB/s/core

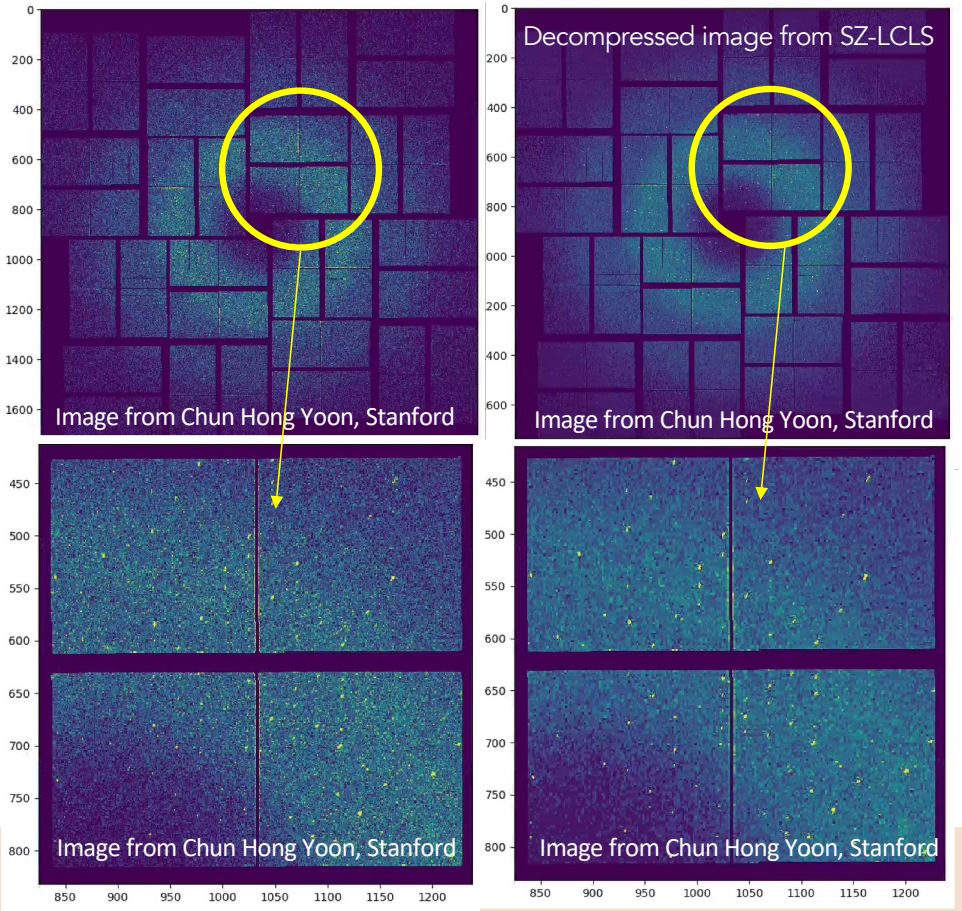
→ True Co-Design (algorithm, hardware)



Experiment cxic0415
run 90 event 104859

Image from the detector

Compression ratio: 30



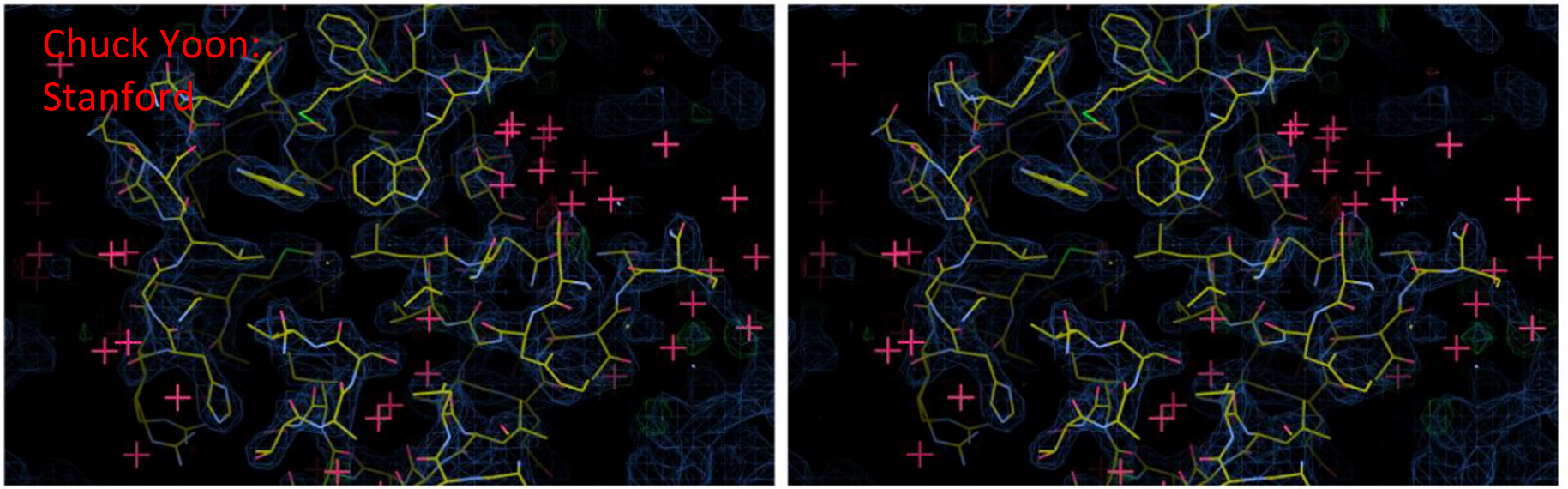
Reducing Streaming Intensity 1/3



Lysozyme on a jungfrau4m detector

Reconstruction of Electron Densities Lysozyme

Very important role in our immune system: breaks up (digests) components of the cell walls of bacteria.



(a) original (b) roibin-sz

The data on the right is 196x smaller (or 631× if also using Non-Hit Rejection)



Questions?